Exact values and certified lower bounds for the no-three-in-line problem in the cube

Aleksei Kudriashov (Alex Komang)
Nusa Dua, Bali, Indonesia

August 2026 — version 0.1 (draft)

Abstract

Let $a(n)$ be the largest number of points of the grid $\{0, 1, \ldots, n-1\}^3$ no three of which are collinear. Pór and Wood proved $a(n) = \Theta(n^2)$; no exact values appear to have been recorded. We determine $a(n)$ for $n \leq 5$ by exhaustive search, certify lower bounds for $6 \leq n \leq 8$, and give an elementary construction showing $a(p) \geq p^2$ for every prime p. Every exact value was computed independently by two programs written to different designs, and every witness is re-verified by testing all triples in exact integer arithmetic.

1 Definitions and elementary bounds

Points of $G_n := \{0, \ldots, n-1\}^3$; a set is *lawful* if no three of its points are collinear in $\mathbb{R}^3$ (equivalently, all 2×2 minors of the difference matrix vanish for no triple). Each of the $3n^2$ axis-parallel lines carries at most two points of a lawful set, and each point lies on three of them, so $|S| \leq 2n^2$; this is the trivial upper bound, attained at $n = 2$.

Proposition 1. *For every prime p, $a(p) \geq p^2$.*

Proof. Let $Q(x, y) = x^2 - dy^2$ with d a quadratic non-residue modulo p, so that $Q(a, b) \equiv 0$ only for $a \equiv b \equiv 0$ (an anisotropic binary form; one exists for every p). Put $S = \{(x, y, Q(x, y) \bmod p) : 0 \leq x, y < p\}$, of size p^2. Suppose three of its points are collinear. Their projections to the (x, y)-plane are distinct (the third coordinate is a function of the first two), hence lie on a line with primitive integer direction (a, b), and the three points are $(x_0 + t_i a,\ y_0 + t_i b,\ \cdot\,)$ with distinct integers t_i, $|t_i| < p$. Along this line the third coordinate satisfies $z \equiv Q(x_0 + ta, y_0 + tb) = Q(a, b)\, t^2 + \ell(t) \pmod{p}$ with ℓ affine, whereas collinearity in $\mathbb{R}^3$ forces z to be an affine function of t. A quadratic congruence agreeing with an affine one at three points distinct modulo p has vanishing leading coefficient, so $Q(a, b) \equiv 0$ and therefore $(a, b) = (0, 0)$, a contradiction. □

Neither $x^2 + y^2$ (anisotropic iff $p \equiv 3 \bmod 4$) nor $x^2 + xy + y^2$ (iff $p \equiv 2 \bmod 3$) works for all p; the choice $x^2 - dy^2$ does. The bound of Proposition 1 is not sharp: $a(5) \geq 40 > 25$ and $a(7) \geq 72 > 49$.

2 Exact values

n	$a(n)$	$a(n)/n^2$	status
1	1	—	exhaustive
2	8	2.00	exhaustive (meets the trivial bound)
3	16	1.78	exhaustive
4	28	1.75	exhaustive
5	40	1.60	*lower bound certified; optimality in progress*

The exhaustive values were produced by two independent programs: a depth-first search over cells with per-line counters and column pruning, and an include/exclude search using, for every pair of cells, a precomputed mask of all cells collinear with them. Their node counts differ by a factor of about 72 ($17/3855/4.21 \cdot 10^7$ against $17/20068/3.03 \cdot 10^9$ for $n = 2, 3, 4$), so the agreement of the values is not an artefact of a shared design. For $n = 5$ the search is split into 256 independent prefixes (the subsets of size ≤ 2 taken in each of the first two z-columns; three points in a column would be collinear, so the prefixes cover the space without omission) and the completeness of the covering is checked mechanically before any claim of optimality.

3 Certified lower bounds

n	$a(n) \geq$	$/n^2$
6	64	1.78
7	72	1.47
8	90	1.41

Obtained by randomised greedy search with destroy-and-repair; the value at $n = 7$ agrees with an independent CP-SAT run. Every configuration is re-verified by testing all $\binom{|S|}{3}$ triples with integer cross products. A dedicated exhaustive proof of optimality is out of reach for $n \geq 6$: a calibrated tree-size estimator (Knuth's random-probe method, validated against the true counts at $n = 4$ to within 4%) gives about $4.6 \cdot 10^{22}$ nodes for $n = 6$.

4 Reproducibility

All programs, journals, witnesses and the coverage checker are in the repository `https://github.com/iwasborninbali/saturation`.

AI/tool-use disclosure

The searches, the programs and the text were produced by autonomous AI agents (Anthropic Claude) under the direction of the author of record, who posed the question, chose what to compute, decided what to claim and takes responsibility for the content. Two agents worked with deliberately different program designs and checked each other's results; no value is reported here that was not obtained twice.